



Dynamic analysis of additively manufactured tensegrity structures

Keivan Davami^{a,*}, Russell Rowe^{a,1}, Ben Gulledge^a, Jesse Park^a, Ali Beheshti^b, Anthony Palazotto^c, Fariborz Tavangarian^d, Sadie Beck^a

^a Department of Mechanical Engineering, University of Alabama, Tuscaloosa, USA

^b Department of Mechanical Engineering, George Mason University, Fairfax, VA, USA

^c Department of Aeronautics and Astronautics, Air Force Institute of Technology, Wright-Patterson AFB, Dayton, OH, USA

^d Mechanical Engineering Program, School of Science, Engineering and Technology, Penn State Harrisburg, Middletown, PA, 17057, USA

ARTICLE INFO

Keywords:

Tensegrity

Vat photopolymerization

SHPB

Energy absorption

ABSTRACT

Herein, we present an analysis, design, and experimental testing of modular prestressed pin-jointed structures constructed from bistable units and inspired by the classical triangular tensegrity prism. Tensegrity structures, characterized by a combination of tension members (cables) and compression members (bars) in a self-equilibrated state, have gained significant attention in engineering over the past two decades due to their unique nonlinear mechanical behavior. The discontinuity of the compression members in tensegrity structures leads to a slightly different failure behavior compared to their lattice structure counterparts, with unprecedented applications. However, traditional fabrication and assembly methods have posed challenges for their widespread adoption. This research benefits from a recently introduced innovative approach for designing and fabricating bistable “tensegrity-like” units where there exists no flexible element in its structure. Vat photopolymerization technology was utilized to create compliant mechanisms based on a triangular tensegrity prism. Two types of structures, namely a double tensegrity-like unit cell and a lattice structure incorporating the triangular tensegrity prism, were fabricated. Quasi-static compression tests were conducted along with high strain rate experiments were conducted using a specialized direct-impact Hopkinson pressure bar setup. Compression tests at both low and high strain rates confirmed the reliable activation of the designed bistable twisting mechanism, even under large displacements, without the need for self-stress. Experimental results at low and high strain rates demonstrated that these low-density units (designed relative density of 20 %) with bistable characteristics are well-suited for applications requiring highly customizable multistable metamaterials. Here, a sudden transition event (snapping), was clearly seen in both quasi-static and dynamic tests. This, indeed, shifts the structure into a secondary stable configuration while maintaining the twisting mechanism throughout the loading cycles. Overall, this study presents a promising avenue for the design and application of tensegrity-like units in various engineering contexts, demonstrating the effectiveness of tensegrities to carry external loads across a wide range of strain rates.

1. Introduction

Mechanical metamaterials, a sub-class of cellular materials, are precisely and meticulously engineered structures with tightly controlled geometries and unusual apparent mechanical performance. Unique properties such as outstanding strength-to-weight and stiffness-to-weight ratios [1–6], frequency bandgaps [7,8], zero-stiffness metamaterials that can isolate vibrating mechanical energy [9], negative overall elastic moduli [10], negative mass density [11–13], auxeticity

[14], and acoustic wave transmission with expected bandgaps or propagation paths [8,15–18] are only some of the reported unusual properties. These attributes are not found in natural materials and are mainly derived from periodic geometries at macro-, micro-, and even nano-scales with intricate patterns rather than the materials themselves. Truss-like structures are the subset of mechanical metamaterials with the underlying material forming an interconnected periodic framework or a continuous plate that is typically corrugated [19]. They have found applications in a wide range of fields, including aerospace, civil

* Corresponding author.

E-mail address: kdavami@eng.ua.edu (K. Davami).

¹ These authors contributed equally.

engineering, robotics, and materials science, with the potential to revolutionize industries by offering new levels of control and performance in mechanical systems.

Tensegrity structures, a newly introduced type of metamaterials, consist of self-stressed pin-connected frameworks composed of continuous cables (strings) in tension and discrete bars (struts) in compression and tension [20]. They exhibit complex nonlinear responses depending on geometry and self-stress levels [21–23]. Tensegrity structures retain their configuration in a state of static equilibrium by upholding a balance between the compressive forces in the bars and the tensile forces in the cables. While it is not intuitively obvious how the loads are transferred in tensegrity structures, they are, indeed, stable systems as a new paradigm in the hands of designers [24]. The stability is defined as the return to the initial configuration after being released from any small disturbance. This stability is present if the locally minimum total potential energy, or strain energy in the absence of external loads is maintained [25]. The shape of tensegrity structures can be changed dramatically with a small change in the potential energy. The particular design of this type of metamaterials makes them light, strong, and deployable [26]. While tensegrity structures have still not been fully introduced to the realm of industry, they have found applications in architectures [27,28], civil engineering [29], mechanical engineering [30,31], and robotics [32,33] due to their lightweight, deployable, and impact-resistant nature. The design and fabrication of tensegrity structures are both challenging, however, with continuous progress in advanced design tool and manufacturing methods, many of these challenges have been overcome. In the territory of design, in general, there are two main groups of form finding techniques namely, kinematical methods and statical methods. Kinematical methods are typically recommended for optimizing the details of structures that are already known [34]. They imitate the motion process of tensegrity structures, and subsequently determine the design of elements to limit the motions, either by adjusting the bar lengths while the cable lengths are kept constant, or by minimizing the cable length, while the bar length is unchanged. A few examples of kinematical methods include the analytical method [35], nonlinear programming [36], and dynamic relaxation [37,38]. On the other hand, and as it can be comprehended from their names, statical methods rely on the principle of statics, where equilibrium equations are used to obtain a self-equilibrated configuration for a specific topology with prestressed members. Subsequently, additional elements are used to add further constraints and minimize the potential energy of the structure [34,39]. A variety of statistical methods have been presented in the literature including force density method [40], energy method [41,42], and reduced coordinates [34]. Tibet et al. presented a detailed comparison of these techniques in [34].

One of the practical predicaments for the fabrication of tensegrity metamaterials is the need for self-stressed tension-only elements. Using advanced additive manufacturing techniques bars can be readily additively manufactured, unfortunately it is currently not technologically possible to create non-rigid strings. This problem has been rectified to a very good extent by introducing conventional lattices with “tensegrity-like” architectures having the capability to mimic the nonlinear response of the corresponding tensegrity structures while conveniently consisting of only struts [20,43]. Similar additively manufactured structures without any actual flexible members have been reported before and referred to as “tensegrity-inspired” structures with the comparable strain energy capacity and compressive response as a buckling tensegrity structure [44]. This innovative solution, along with the implementation of emerging additive manufacturing techniques have resulted in fabricating features mimicking the nonlinear response of tensegrity structures with bistable behavior and other unique characteristics [43–46]. The main shortcoming of this strategy might be the introduction of bending stresses yet, they can be significantly reduced by tapering the bars at the ends, or by using flexible hinges [20,47,48]. Multiple research groups have employed this technique with intriguing findings. For instance, a reduction in the response nonlinearities of

tensegrity lattices has been reported which was attributed to the bending-stiff nodal connections in additively manufactured tensegrity structures [49]. There were also reports on combining additively manufacturing methods and traditional techniques. For example, Amendola et al. [50] used an electron beam melting technique to fabricate tensegrity prisms out of titanium alloy. Subsequently, cross-strings were added to the structures with internal prestress. It was shown that under compressive loads, the structures showed a stiffening response.

The mechanical behavior of tensegrity structures depends on connectivity, geometry, material properties, imperfections, and prestress of the constituent elements, among others [46,51,52]. For example, a tensegrity prism, a three-dimensional self-stressing system with both tension and compression members [53], can exhibit either stiffening or softening responses under compressive loads, depending on the design. Jianguo et al. showed that with even minor initial imperfections in compressive member the stiffness of the tensegrity structure is reduced significantly and a strong nonlinearity in both the deformation of the bars and structure stiffness occurs [54]. Rimoli et al. fabricated three-dimensional tensegrity lattices from truncated octahedron elementary cells, and further studied the dynamic response of an $8 \times 8 \times 8$ tensegrity lattice. The results showed that tensegrity-based metamaterials are an ideal choice for energy absorption thanks to their significant strength at low density, their resiliency under large deformation, and the way they disperse and exhibit asymmetry in wave propagation [55]. Fraternali et al. studied the softening behavior of tensegrity structures under compressive loads and large displacements [51]. It was shown that while thick prisms depict an elastic softening behavior in compression under low prestrains, slender structures require a high prestress in order to illustrate elastic softening when they are under compressive loads. In addition, it was deduced that the response of the structure is contingent upon the aspect ratio of the structure, the magnitude of the applied prestress, and the material properties of the constituent elements. Shan He et al. studied the directional snapping instability in a quadrangular prismatic tensegrity structure when they were under quasi-static compressive loads and two different self-equilibrated stable states were observed [56].

Recent studies show that the tensegrity structure's compliance offer exciting possibilities for enhanced energy absorption efficiently [20,56]. Zhang et al. utilized the buckling of the bars to optimize the energy absorption of the tensegrity structures numerically [57]. It was also demonstrated that there is a cubical relationship between the size and prestress level with the energy absorption. It was suggested that tensegrity structures can mitigate vibration providing a flexibility in the vertical direction and decent horizontal shear stiffness [57]. Micheletti et al. simulated the impact loading on a chain of tensegrity prisms and lumped masses [58]. They indicated that the impact dynamics of these structures involves a combination of thermalization phenomena, in areas adjacent to the impact region, and the generation and transmission of compressive waves ahead of the thermalized zones. Beauer et al. [46] reported orders of magnitude increase in energy absorption without failure in tensegrity structures in comparison with same-strength lattice structures. They also indicated a 25-fold enhancement in deformability in this type of structure, attributed to the delocalized deformation. However, their experiments were conducted at low strain rates and only for small-scale specimens. Davini et al. [52] reported that one could potentially explore a novel utilization of tensegrity structures, where the transmission of energy takes the form of solitary waves, and additionally, localization phenomena like energy entrapment and shock dissolution might manifest. Fraternali et al. [59] numerically investigated the wave dynamics of tensegrity prisms. It was suggested that these systems can serve as ultimate impact mitigation solutions, however, depending on local and global prestress levels, they would require a relatively large number of unit cells, typically on the order of 1000.

Herein, we experimentally demonstrate the quasi-static as well as high strain rate behavior of tensegrity-like lattice structures with highly

nonlinear mechanical responses and only composed of bars (struts) fabricated using a vat photopolymerization system. We focus on the strain rate effect under uniform axial loading. The quasi-static tests were conducted under compression forces while the high strain rate experiments were performed using a direct impact technique. The snapping behavior was observed under both quasi-static and dynamic loads. Ultimately, the structure experienced complete axial collapse as a consequence of the snap buckling phenomenon.

2. Experimental method

2.1. Design

Here, a double-T3 unit cell concept was adopted. A T3 unit is a triangular right prism structure where the edges are pre-stressed cables, only sustaining tensile loading, while diagonal supports are bars sustaining either tensile or compressive loads depending on the form of deformation. A thorough description of the T3 unit cell and the double-T3 unit is given by Intrigila et al. [20]. Fig. 1a shows parameters d , the diameter of the struts and bars, as well as h , the height of the T3 unit cell (lower and upper units). The inset in Fig. 1a shows a single unit tensegrity structure where the circumscribed radius of the lower base is shown with “ a ” and the circumscribed radius of the upper base is shown with “ b ”. The diameter of the outer and middle bases is the same as “ a ” and “ b ”, respectively, in the inset of Fig. 1a. The pre-stress on the cables is a result of the respective rotation between the base and top of the structure, referred to as the twist angle ϕ . The height of the T3 unit can vary but the ϕ should be $\frac{\pi}{6}$ to reach a stable self-stressed configuration. Intrigila et al. found varying ϕ between 15–22° results in a bi-stable behavior [20]. For this study, a ϕ of 19° was chosen as it gave the most uniform response. Fig. 1b shows a tensegrity lattice made of 20 double unit cells, 10 on the top layer and 10 on the bottom. The T3 unit cell herein qualifies as a Class 1 Tensegrity Structure as it features precisely one bar connecting to each node, as specified in [60]. The thickness of the bars and cables, d , was then edited to create a structure with a designed relative density of $\bar{\rho} = 20\%$, to allow for comparison to existing lattice structures (Table 1). A single unit cell at the same size as the lattice structure units would be too weak to register load accurately due to the limitations of the dynamic impact setup in this research. Therefore, it was instead opted for the single unit to be scaled to fit the same impact area of our dynamic impact setup.

The load applied on the structure is distributed across the 3 nodes in the top T3 cell and propagated to the lower T3 cell. The straight red arrows shown in Fig. 2a illustrate the compressive loading direction while the curved arrow indicates that the structure rotates clockwise. Fig. 2b shows the bottom cell is a reflection of the top cell, therefore the rotation of the middle nodes caused by deformation of each T3 unit cell is complimentary and therefore purely translational. Additionally, Fig. 2b has been color coded so that the bars are thick and blue, and the cables are thin and red to display which members experience

Table 1
Geometric dimensions in Fig. 1.

Parameters	Single Unit	Lattice
ϕ (degrees)	19	19
a (mm)	30.35	9.06
b (mm)	26.30	7.85
h (mm)	22.75	6.80
d (mm)	3.35	1.00

compressive and tensile loads or only tensile loads, respectively. It needs to be noted that the cables and bars in the structures in the current study had the same diameter (Fig. 1). The nodes at the top and bottom of the double unit cell T3 structure are aligned vertically, and the middle row of nodes has a relative twist of 19 degrees, indicated in Fig. 2c, which is required for a T3 structure to exhibit a multi-stable compliant mechanism.

2.2. Fabrication

The digital design of the specimen was generated using SOLIDWORKS, followed by slicing through PreForm software. The resultant file was then sent to the Form 2 SLA 3d printer for sample fabrication. Samples were printed using a commercial resin (Tough 2000 resin) with a 50 μm layer thickness, vertical sample orientation (print layers perpendicular to direction of loading), and utilizing a raft to allow for easy sample removal. Next, the specimens were washed for 20 minutes in isopropyl alcohol to remove uncured resin and then removed to air dry. After the wash, the specimens were exposed to UV light in a high temperature environment for post-curing. In order to achieve the best mechanical properties of the resin, the specimens were post-cured for 60 minutes at 60 degrees Celsius. The schematic of the printing process is shown in Fig. 3.

2.3. Material Characterization

In order to characterize the base material, five Type IV tensile specimens were manufactured as outlined by ASTM D638 – 22 with a thickness of 2 mm (pictured in Fig. 4). Tensile tests were performed on the specimens using a MARK-10 ESM 303 motorized force testing machine with a Series 4 load cell. Series 4 digital force gauges are designed for tension and compression force testing in numerous applications across a wide range of industries, with capacities from 0.12 lb to 500 lb (0.5 N to 2500 N). The gauges feature a sampling rate of 3,000 Hz, producing precise results under a wide range of test conditions. Accuracy is $\pm 0.2\%$ of full scale, and resolution is 1/2000. The tests were carried out with a constant displacement of 13 mm/min (a strain rate of 0.0085 s⁻¹) and a data sampling rate of 20 Hz.

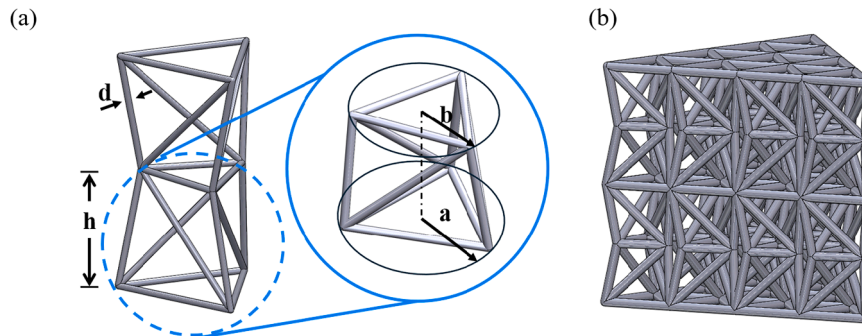


Fig. 1. Geometry of the designed (a) double-T3 unit and (b) T3 lattice unit tested in this study. The inset in (a) shows the T3 single unit as the constituent structure that has been repeated and formed the two structures tested in this research.

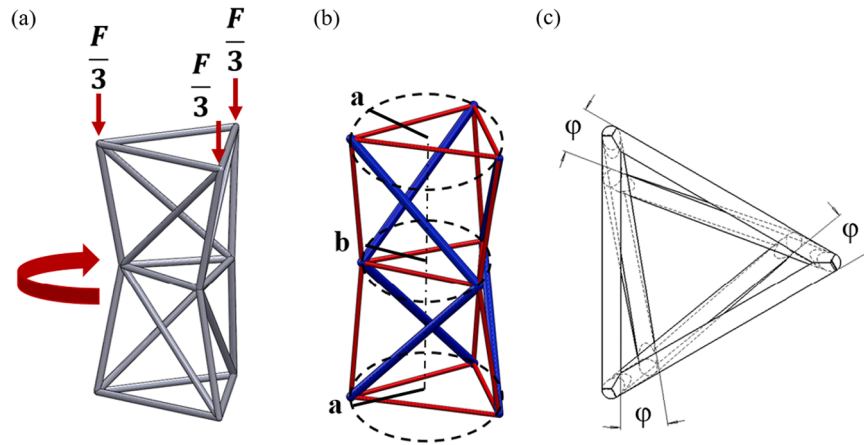


Fig. 2. (a) A compressive load leads to clockwise rotation of the double tensegrity unit, (b) where bars are represented by thicker blue lines, and cables are represented by thin red lines. (c) The resulting twist angle in the double-T3 unit seen from a top-down view.

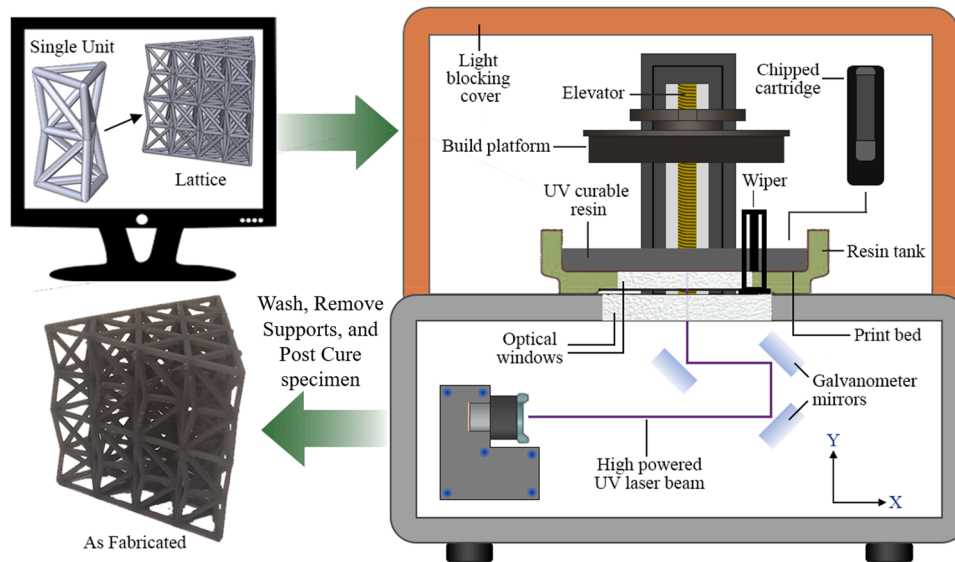


Fig. 3. The fabrication steps of the structures using a vat photopolymerization system.

2.4. Compression tests

To compare the behavior of the single unit cell and lattice tensegrity structure, quasi-static uniaxial compression tests were performed using the same MARK-10 ESM testing machine mentioned above. For quasi-static uniaxial compression loading to assess the mechanical behavior of both the single unit cell and the lattice tensegrity structures, a MARK-10 ESM testing machine was used (the same was used for the tensile loading). The compression load was applied at a controlled displacement rate of 13 mm/min, corresponding to a strain rate of approximately 0.0045 s^{-1} for the double-T3 unit and 0.0076 s^{-1} for the lattice structure with data recorded at a sampling rate of 20 Hz. For both quasi-static and direct impact testing, the specimens were placed between parallel compression platens, ensuring alignment to maintain uniaxial loading conditions. Force-displacement curves were obtained and used for the interpretation of the mechanical response of the structures.

2.5. Direct impact tests

Direct impact tests were performed on both single unit cell and lattice tensegrity structure using a direct impact setup built on an REL Inc. (USA) split Hopkinson pressure bar (SHPB) with an aluminum

cylindrical bar with a length of 304.8-mm and a diameter of 38.1 mm applying uniaxial loading to the specimen. It is known that insufficient lubrication at the interface can lead to unreliable dynamic compressive strength due to frictional effects. To remediate this, Super Lube® Synthetic Grease with Syncolon® was applied to the interface between the compression specimen and the platens. The thin layer of lubricant significantly reduces the friction. Impact tests were performed using gas at a pressure of 78.9 kPa corresponding to a $\sim 6.5 \text{ m/s}$ exit velocity (strain rate of $\sim 133.6349 \text{ s}^{-1}$ for the double-T3 unit and 226.4808 s^{-1} for the lattice structure). Displacement data was collected using target tracking on a video captured via a Vision Research Phantom V611 high speed camera recording at 55,000 FPS. While there are no external constraints specifically designed to prevent lateral movement or rotation of the specimen, the design and setup of the test ensured that uniaxial compression was imposed. The rigid, flat, and parallel compression platens were carefully aligned with the specimen, and the inherent stiffness of the testing machine further supported this alignment. This setup ensured that the end faces of the specimen remained parallel throughout the test, thereby maintaining uniaxial loading conditions. A schematic of the direct impact setup is shown in Fig. 5.

Direct impact testing closely simulates real-world impact conditions, making it ideal for evaluating material performance in scenarios like

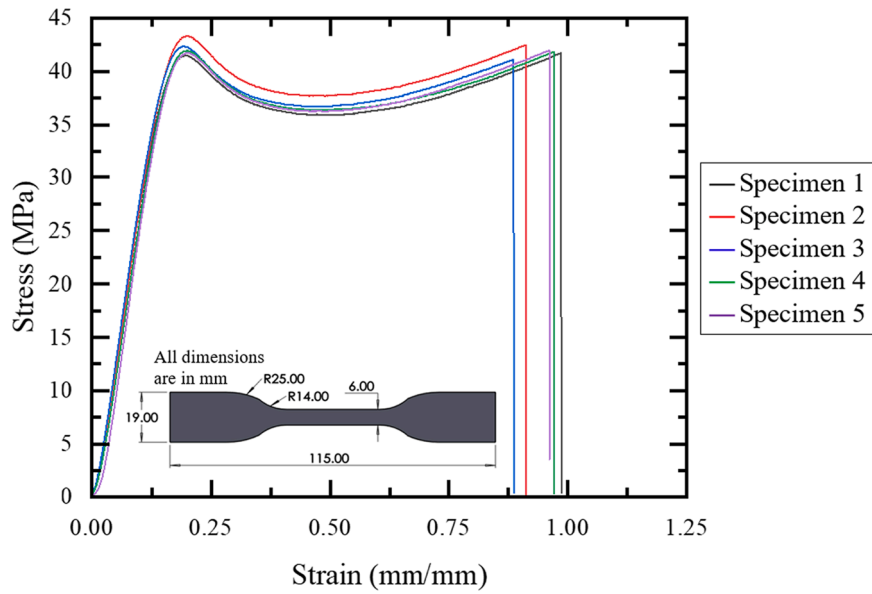


Fig. 4. Tensile stress-strain curves of the resin Tough 2000.

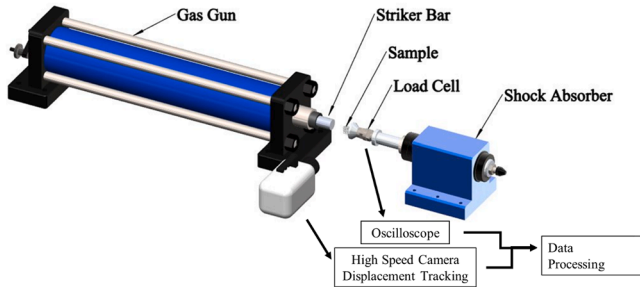


Fig. 5. A schematic of the direct impact setup.

crashes or ballistic impacts. It captures complex, high-energy interactions that may be difficult to replicate with the more controlled but less realistic Hopkinson bar test. Direct impact testing involves launching a projectile at high speeds to directly impact a specimen, simulating real-world conditions such as ballistic impacts or crash scenarios. This method, while closely mimicking actual impact events, offers less control over the strain rate, which is more variable and dependent on the projectile's velocity and material properties. In contrast, the Hopkinson bar test provides a highly controlled environment where a specimen is sandwiched between two bars, and a stress wave is induced by striking one of the bars. This setup allows for precise adjustment and measurement of strain rates, yielding accurate stress-strain curves ideal for studying the dynamic behavior of materials. However, while SHPB testing excels in generating well-defined and repeatable data, it may not replicate the complex, real-world impact conditions as effectively as direct impact testing.

In the direct impact test, the response of the material to the impact is recorded using various sensors and instruments, such as high-speed cameras, load cells, and strain gauges. These devices capture data like the force of the impact, the displacement or deformation of the specimen, and the time duration of the event. In this experiment, we used a load cell and a high-speed camera to record the data. The load cell has a 0-120kN range with a calibration range of 0-12kN and 0-120kN with a $\pm 0.4\%$ linearity within the calibration range.

3. Results and discussion

3.1. Quasi-static testing

3.1.1. Mechanical Properties of the Constituent Material

The stress-strain curves of the constituent material are shown in Fig. 4 for the five different ASTM tensile specimens with the same geometry. As seen, under uniaxial tensile loading, the parent material shows ductile behavior. While there was a very slight difference in the fracture strain of the tested specimens, other mechanical properties such as Young's modulus (~ 295.32 MPa), tensile strength (~ 42.13 MPa) and strain at yield (~ 0.195 mm/mm) were very close for all specimens, indicating the repeatability of the measurements.

The resultant mechanical properties of the specimens are summarized in Table 2. The data shows high repeatability ensuring consistency of the measurements.

3.1.2. Mechanical Properties of the tensegrity Structures

Figs. 6a and 6b show the quasi-static compression results of a single unit cell as well as a lattice structure associating with the pictures illustrating the deformation trends for each case. As seen, the quasi-static compression of the tensegrity-like unit structure shows a non-linear behavior. The stress first peaks when the single unit cell structure reaches its elastic limit (Point I in Fig. 6a). There is an increase in the force up to a value of ~ 500 N at a displacement of ~ 4 mm up to this point where the tensegrity-like structure shows a stable behavior. After Point I, the behavior becomes unstable where the stress diminishes as a result of the lattice bars buckling. The compressive limit load is followed by a subsequent drop in the force where the structure becomes unstable for a short amount of time (Point I to Point II) but with an increase in the displacement beyond ~ 7.5 mm (Point II) a level of stability is achieved

Table 2

Results of 5 tensile tests at a rate of 13 mm/min (a strain rate of 0.0085 s $^{-1}$).

Specimen	Tensile Peak	Strain at Peak	Young's Modulus
#	(MPa)	(%)	(MPa)
1	41.5	13.29	250
2	43.3	19.54	268
3	42.3	19.02	258
4	52.0	19.93	263
5	41.7	19.67	275

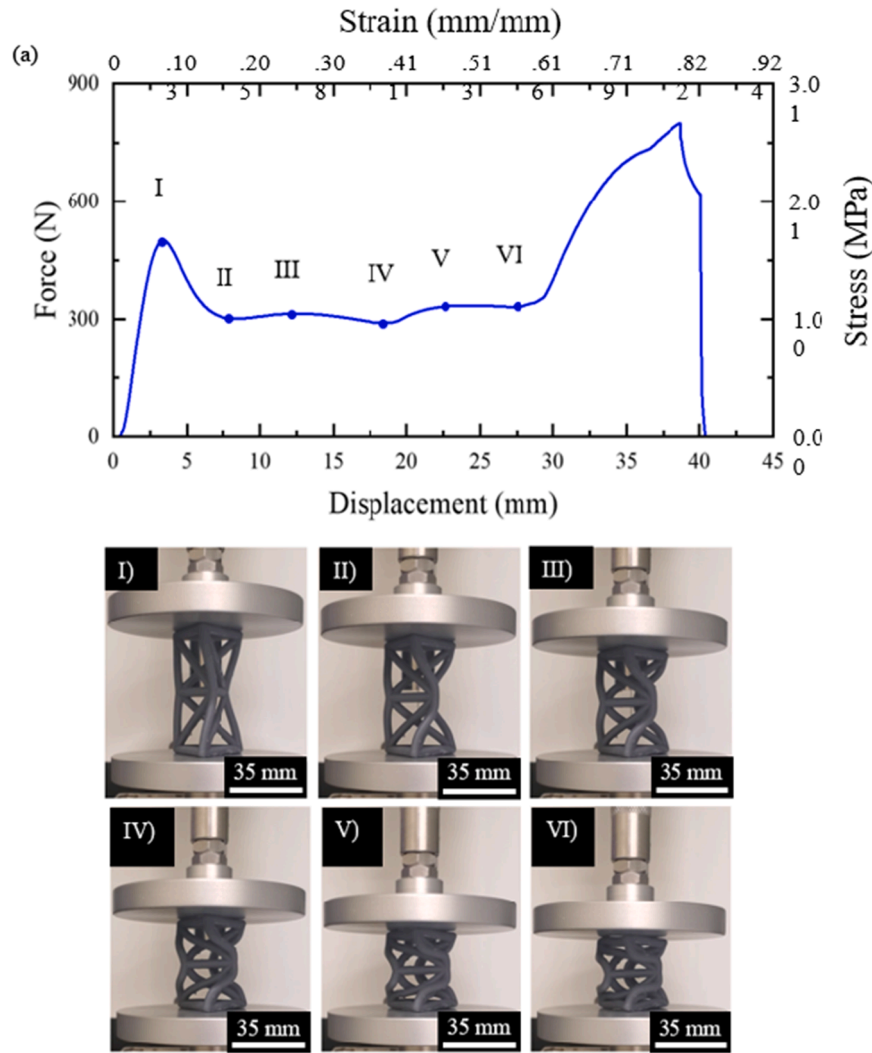


Fig. 6. Load-displacement curves obtained from quasi-static compression tests conducted on a double-T3 tensegrity-like structure. (a) A unit cell, and (b) a 3×3 lattice structure. Beneath each figure, selected screenshots depicting different stages of deformation are presented associated with the force-displacement curves, denoted with Greek numbers. The video clips of both of these tests are available in the supplementary information.

where the force stays at around 300 N without any significant changes (Point II to Point VI). Between Points II and VI, the structure has a relatively stable force-displacement response, with a continuous slight stress increase, before densification that starts at 28 mm of displacement. After visually examining the images, Fig. 6a (II to VI) and the testing video (Video S1), it becomes apparent that the structure's response is irregular, as the central base of the unit deviates from the horizontal plane slightly. This suggests that the two prisms making up the structure may exhibit distinct responses. The buckling of the members is clearly seen in the video (Video S1) and the counterclockwise rotation about the vertical axis of the middle triangle is also seen in the recoded video clip (Video S2). In this region, the substantial twisting of the middle triangle necessitates significant deformation of the compliant hinges, brought about by the alteration in angle between the adjacent bars. When the densification starts (Point VI), the applied force rapidly goes up and reaches a value of almost 850 N, before the structure completely fails.

Fig. 6b shows the force-displacement plot for a lattice composed of two layers of ten units each. The force-displacement responses of the lattice structure compared to the unit cell are largely unstable, where increasing displacement often causing major negative stiffness events, indicating a sudden decrease of the structures' load bearing capability.

Similar to the single cell unit, the force-displacement diagrams illustrate that the stress reaches its maximum at the elastic threshold of the lattice structure (Point I in Fig. 6b). Following this stage, the sustained load declines as a result of the upper bars undergoing buckling. The extent to which load drops is approximately 45% higher for the unit cell, 494 to 294 N, than for the lattice structure, 219 to 131.5 N. The structure is unstable and the load drops, until Point II where the structure starts to show a secondary stable position. With an increase in the displacement, the load increases again and reaches another peak (Point IV), before repeating the same behavior again where the stress drops until Point V. At the end, where the densification happens, there is a significant increase in the applied load (Point VI and slightly after). The two-snapping event that occurs in this structure (one from Point I to Point II, and the other from Point IV to Point V) are each attributed to one layer of the structure. The corresponding snapshots from the video clips (Video S1) are also shown below the graph in Fig. 6b. Vangelatos et al [43] compressed similar structure but in microscale under quasi-static loads and observed a similar behavior. They attributed this two-stage rotation to an increased stiffness in the lower layer, which, in turn, might be linked to the lower layer's connection to the fabrication plate and/or the presence of unintended polymerized regions within the lower layer resulting in a different stiffness. However, nanoindentation tests on

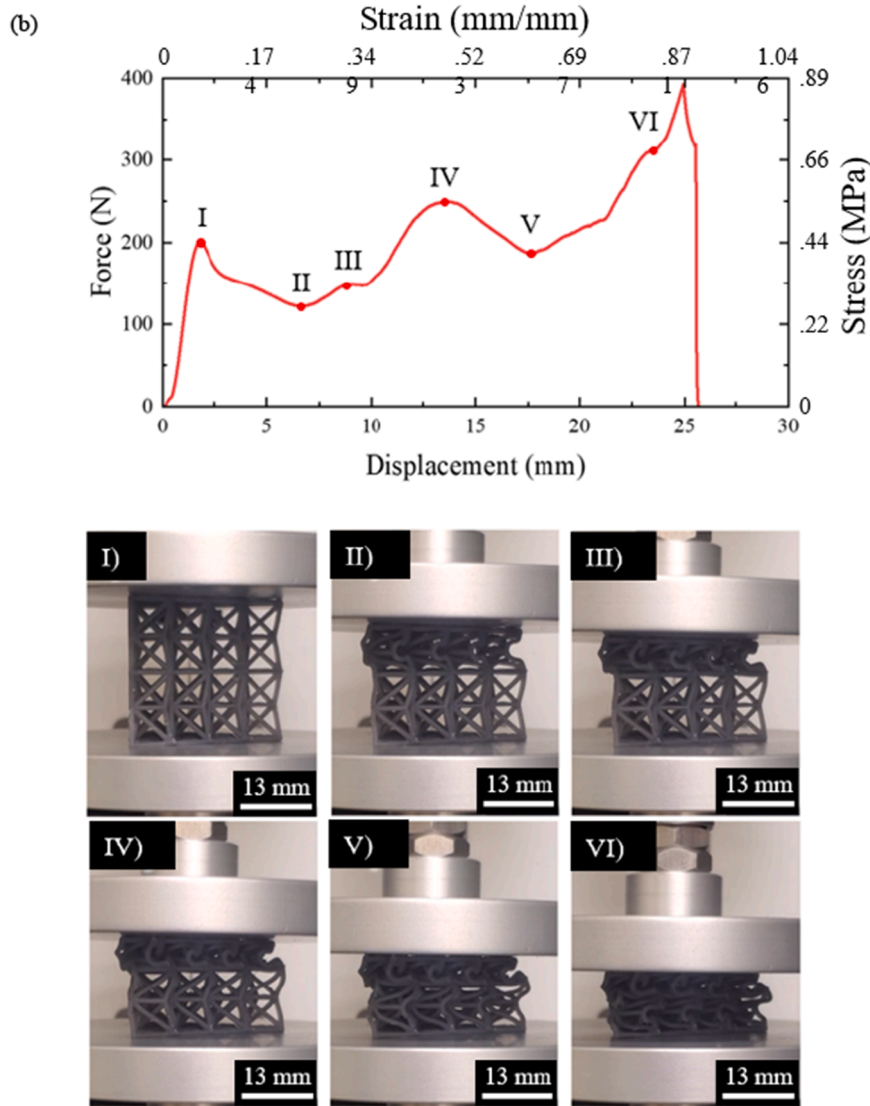


Fig. 6. (continued).

different faces of a cylindrical structure showed fairly uniform stiffness for our case. Upon inspection of the quasi-static compression images as well as the movie of the compression experiment, the discernible rotation of the middle triangles is revealed. Additionally, it is important to highlight that while there is an obvious fluctuation in the stress response of the lattice structure between Point I and VI, such a shift is not observed in the unit cell structure. This is due to the fact that buckling of the bars happens more uniformly for the unit cell specimen, while it is more localized and develops progressively for the lattice specimens, causing a sudden drop in the force during the compression. At least 5 tests were performed for each static experiment, where Fig. 6 shows representative graphs. Similar behaviors were observed for all the tests, however, to quantitatively demonstrate the variability among them, initial peak load as well as the specific energy absorption (SEA) (see

section 4.3) were estimated from all the tests and their associated average and maximum/minimum values are listed in Tables 3 and 5.

Fig. 7 shows the optical microscopy of the bars after the quasi-static compression tests. In tensegrity structures, the cables only experience tensile loads. The bars typically fail due to compressive stresses as well as shear stresses upon compression. Here, the majority of the fractures can be seen close to the nodes. Typically, the fractured faces have a 45° angle. The resulting stress state in the beam sections is much too complicated to be reduced to only one state. It is assumed that a combination of different types of stresses are responsible for the failure of the structure. Many of the rods' failure is attributed to the buckling (Video S1). It has been reported that the initiation of buckling is a good indicator of the tensegrity structure failure. In fact, it is crucial for the bars to remain in the elastic buckling area in order for the structure to have optimized impact behavior along with enhanced strength to weight ratio [61].

Scanning electron microscopy (SEM) images of the fractured bars can be seen in Fig. 8. Here, common defects in additively manufactured structures such surface imperfections are present in the specimens. Yet, it does not seem that the bars' fracture was affected significantly by these defects, nor the fracture initiated from the defects in the quasi-static loading. Most of the fractures occur near the nodal intersection of multiple cables and bars.

Table 3

Load at the initial peak of the lattice structure and unit cell under quasi-static loading.

Structure Type	Load at Initial Peak (N)		
	Average	Max.	Min.
Lattice Structure	208.3	219.0	200.5
Single Unit	484.8	500.0	460.5

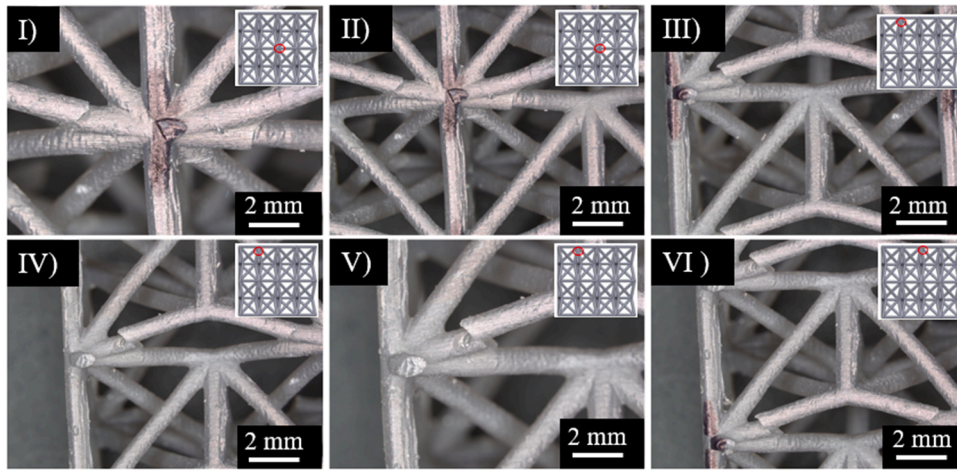


Fig. 7. Optical microscopy images of the fractured structures after the quasi-static compression tests.

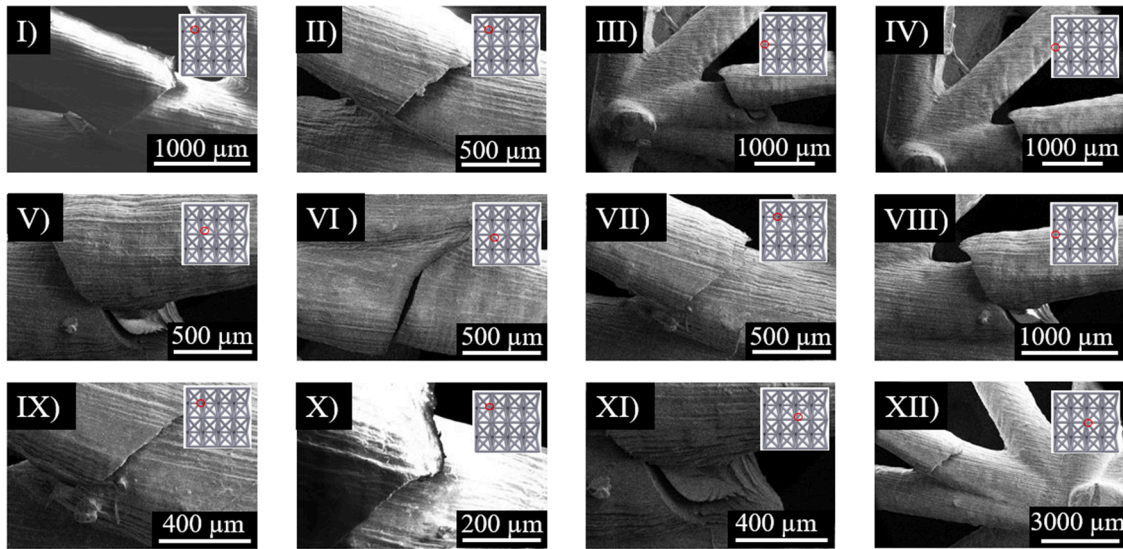


Fig. 8. Scanning electron microscopy images of the fractured structures after the quasi-static compression tests.

3.2. Dynamic testing

To assess the response of the unit cell under high strain rates, we conducted direct impact tests under an impact speed of 6.7 m/s. The structures maintained their multi-stable response with the two layers of the lattice array failing independently matching the results seen in our quasi-static compression tests but with much less stability. In Fig. 9, a series of snapshots from the high strain rate test is observed for the unit cell. To track deformation, the red circular marker remains fixed at the node starting location for all images, while the blue marker follows the node displacement. Notably, the rotation of the middle triangle in these snapshots is observed (Video S3). Also, Fig. 9 presents the force-displacement graph for a representative unit cell. Unlike the behavior of the structure under quasi-static loads, the specimens show highly localized deformation and unstable force-displacement responses under the high strain rate, with pronounced failure-inducing force drops. Immediately following the impact, there is a rapid increase in force up to 750 N at around 2 mm (Point III), which is followed by a sharp drop to approximately 500 N (Point IV) and then relatively steady with small oscillations in the force magnitude. These oscillations can be attributed to the stress wave traveling back and forth within the structure, vibrations in the structures, and rotation of the structure's faces during compression, as detailed in Pajunen et al.'s work [44]. Buckling of the

bars initiates within this range where at around 14 mm of displacement (Point VI), the bars fail due to buckling, leading to the failure of the right side of the unit cell. The remaining part of the structure withstands the force until it reaches approximately 400 N (VIII), at which the bars of the unit cell fail again, resulting in a sharp drop in the force at a displacement of approximately 25 mm (IX). At this displacement, the structure experiences complete failure, while the densification of the specimen causes a final significant increase in the force value.

Fig. 10 shows snapshots from the high strain rate test of the lattice structure. Once again, in this figure, the red circular marker remains fixed, while the blue one follows the unit's deformation. The full video is available in Video S4. Initially, the rotation of the left side of the structure occurs (Fig. 10 (I)-(V)), and with additional displacement, the rotation of the right side of the structure commences and continues until the structure's failure (Fig. 10 (VI)-(IX)). We hypothesize that the load transfer in the structure results in "delocalized deformation" at high strain rate in contrast with other cellular structures where localized deformation leads to catastrophic failure [46]. However, further experimental and theoretical studies are needed to understand the mechanism involved.

The graph in Fig. 10 displays the force-displacement curves for the lattice structure under the impact load. The synchronized localized failure of a significant number of elements within a layer of the

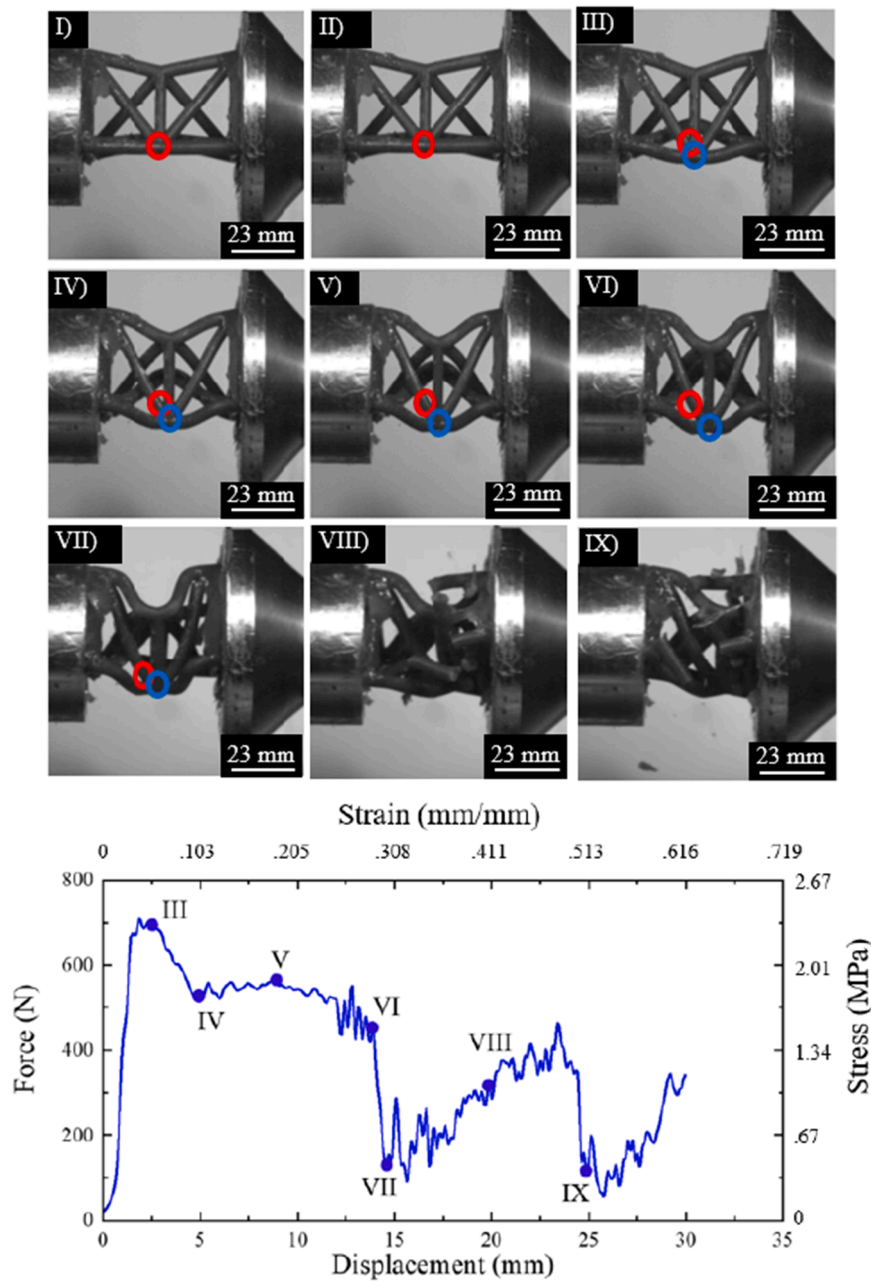


Fig. 9. (I-IX) a series of snapshots from the high strain rate test of the unit cell. In these images, the red circular marker remains fixed, while the blue one follows the unit's deformation. Red and blue markers are not included on VIII and IX due to failure of the structure.

structure, results in an effective negative stiffness response that culminates in catastrophic failure again. Compared to the unit cell, the increase in the load is more gradual upon impact, indicating a different mechanism for load absorption in the lattice structures. It initially rises until it reaches its first peak (depicted as Point IV in Fig. 10), which corresponds to the moment when the structure hits its elastic limit. It needs to be pointed out that herein, the “elastic limit” is an abstract term showing the overall behavior of the structure, and some of the individual constituents of the structure have already gone under a plastic deformation at this point. Subsequently, the force decreases due to the buckling of the bars in the first layer (shown as Point V), and their subsequent fracture, only to rise again and reach a second peak (Point VI) when the bars come into close proximity in the left-side layer and a local densification happens, while the right-hand layer start bearing the majority of the loads. The seesaw pattern that is observed in these force-displacement graphs are attributed to the failure of individual bars

under the compressive loads. After reaching another peak (Point VI), the force again experiences a second drop, attributed to the buckling of the right-layer lattice bars (Point VII), followed by fluctuations due to the fracture of the individual members, and subsequently starts rising up once more as the bottom lattice structure densifies. The force undergoes a significant increase once it starts densification (Point IX).

The significant force differences between the statically and dynamically deformed structures, as seen in Figs 6b and 10 respectively, can be attributed to a combination of dynamic amplification effects, inertial forces, nonlinear structural response, and friction. Dynamic amplification occurs when the rapid application of force under dynamic loading conditions leads to an increase in the effective forces experienced by the structure. This is primarily due to higher inertial effects, which result in localized stress concentrations and an earlier onset of failure mechanisms such as buckling and snap-through. These effects are particularly pronounced in lattice structures, where stress waves propagate and

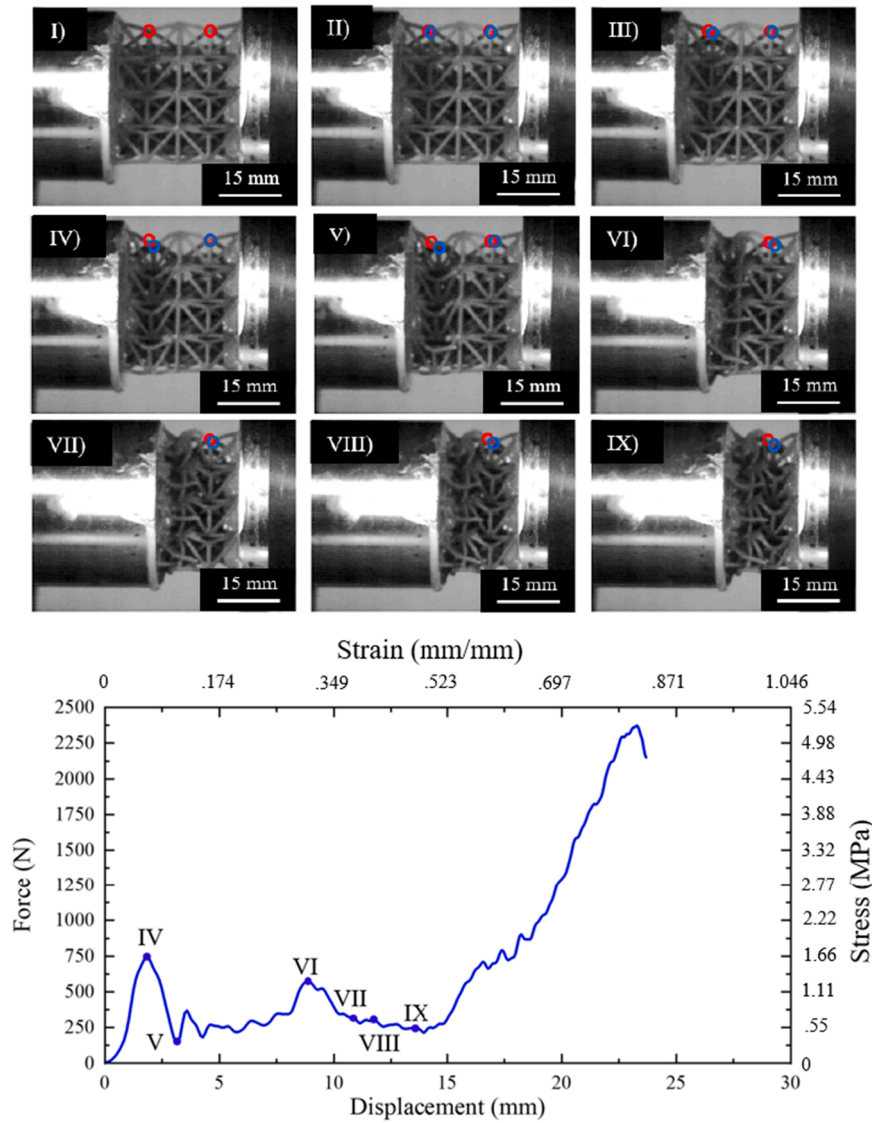


Fig. 10. (I-IX) a series of snapshots from the high strain rate test of the lattice structure along with the force displacement curve. In these images, the red circular marker remains fixed, while the blue one follows the unit's deformation. The images depict the advancement of the failure as it gradually proceeds layer by layer until it reaches a strain level significant enough to completely crush the specimen.

interact within the framework, leading to the significantly higher forces observed in dynamic tests. Additionally, tensegrity structures are particularly susceptible to abrupt changes in force distribution due to factors such as strut collisions and large displacements during dynamic events. The contact force from surface friction arises when strut collisions happen and contacting struts are inclined to slide past each other. Friction is a complex phenomenon that can lead to different behaviors, such as sticking or sliding. In dynamic scenarios, these interactions can cause rapid structural state changes and localized force spikes, further magnifying the forces experienced by the structure. This is consistent with the larger force peaks observed in Fig. 10 compared to Fig. 6b.

Similar to static tests, at least 5 tests were performed for each dynamic experiment with Figs 9 and 10 depicting representative graphs. Relatively consistent behavior was observed among all tests. To further quantify this, initial dynamic peak load as well as specific energy absorption (see section 4.3) were estimated from all the tests where their associated average and maximum/minimum values are listed in Tables 4 and 5.

The lattice structures were tested at different impact velocities, namely, 6.5 m/s (a strain rate of 226.4808 s^{-1}), 7.5 m/s (a strain rate of 261.3240 s^{-1}), and 8.5 m/s (a strain rate of 296.1672 s^{-1}) to evaluate

Table 4

Load at the initial peak of the lattice structure and unit cell under dynamic loading.

Structure Type	Load at Initial Peak (N)		
	Average	Max.	Min.
Lattice Structure	718.5	744.7	679.5
Single Unit	718.0	756.6	688.3

Table 5

SEA values of the lattice structure and unit cell under quasi-static and dynamic strain rates.

Structure Type	Quasi-Static (J/gr)			Dynamic (J/gr)		
	Avg.	Max.	Min.	Avg.	Max.	Min.
Lattice Structure	1.770	1.969	1.599	2.178	2.262	2.027
Single Unit	2.346	2.469	2.102	2.078	2.302	1.961

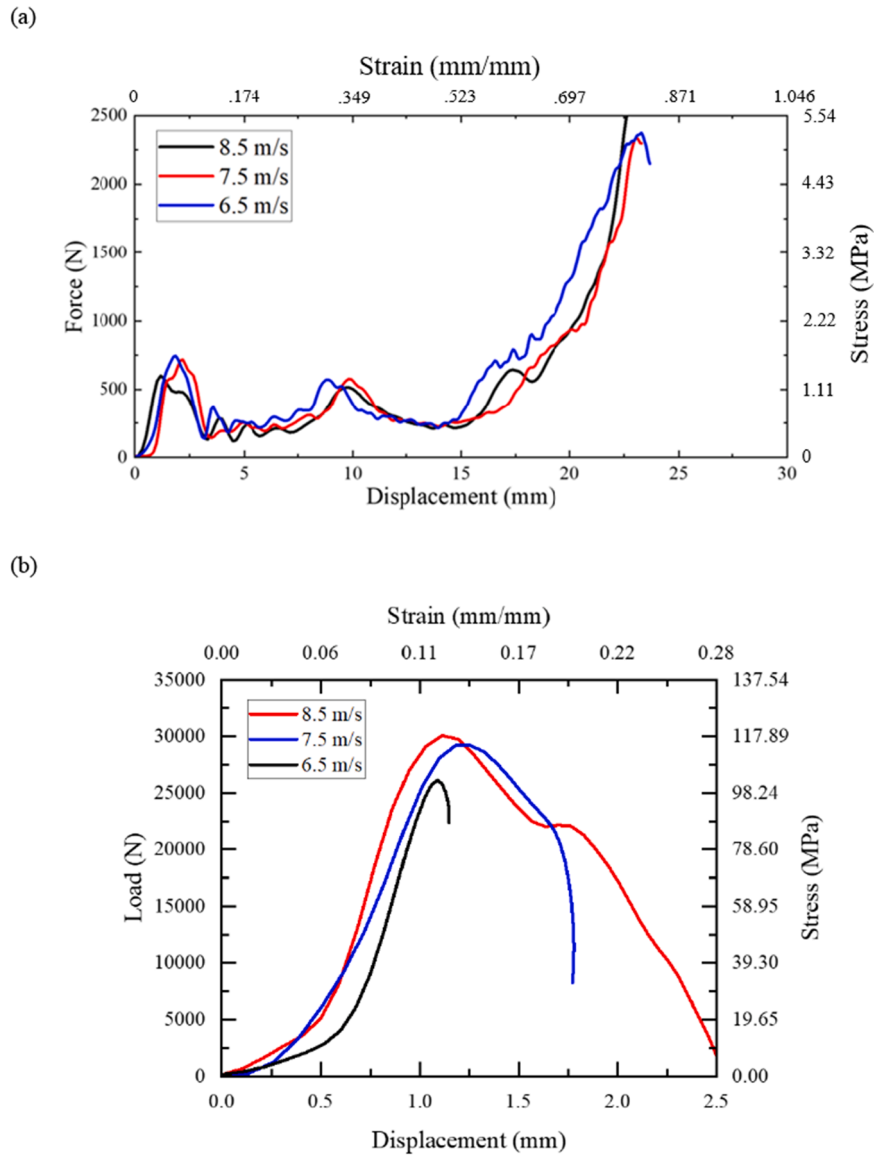


Fig. 11. (a) The force-displacement response of the lattice structure under an impact loading at different velocities, (b) the force-displacement response of a simple cylindrical part under an impact loading at different velocities.

how different velocities affect the response of the structures (Fig. 11a). The specimens showed a relatively similar behavior. The main difference that was observed was in the strain at peak loads that was slightly higher at the lowest strain rates and relatively similar for 7.5 m/s and 8.5 m/s strain rates. No significant strain rate dependency was observed. In order to evaluate if the base material shows a similar behavior under different impact velocities, dynamic tests were conducted on a cylindrical specimen with a diameter of 18 mm and a height of 9 mm at different impact velocities, namely, 6.5 m/s (a strain rate of 722.2222 s^{-1}), 7.5 m/s (a strain rate of 833.3333 s^{-1}), and 8.5 m/s (a strain rate of 944.4444 s^{-1}). The results are presented in Fig. 11b. As can be seen here, the base material also shows a similar behavior and does not show any significant strain rate dependency at the tested velocities.

3.3. Shape recovery

Due to the nature of these structures' design and material, when released from compression, substantial shape recovery was observed in the structures. To assess this capability, a series of quasi-static tests were conducted consecutively, with each specimen given approximately 2

hours to regain its original shape. The structures demonstrated an impressive ability to recover up to $\sim 98\%$ of their full shape after a single loading cycle as seen in Fig. 12. In this figure, the structure was deformed under quasi-static loading at a strain rate of approximately 0.0076 s^{-1} and with a strain of 80%. At the maximum deformation, a load of 300 N was recorded (similar to point VI in Fig. 6). The load was slowly removed and the recovery process at various time intervals, ranging from immediate release ($t=0 \text{ s}$) to a period of two hours ($t=2 \text{ hr}$) was imaged (Fig. 12). As can be seen in these figures, the recovery process is gradual, but highly effective. The process was repeated multiple times, however, after each round of loading, the structure started to lose more strength and reached the same initial deformation with a smaller load value. This can be attributed to the progressive failure of the constituent elements with each additional round of loading. The shape recovery also, started to decrease after the second round of loading and dropped to less than 70% of the original shape after the third round of loading. Further exploration of this shape recoverability property could unveil potential applications for these structures.

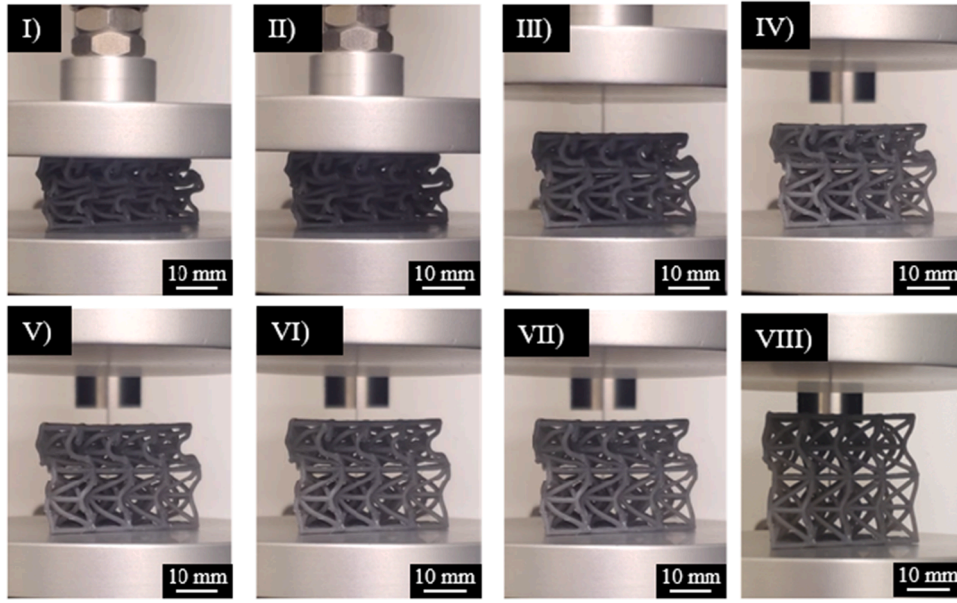


Fig. 12. The shape recovery of the structure after quasi-static testing. (I) $t = 0$ s, (II) $t = 1$ s, (III) $t = 5$ s, (IV) $t = 30$ s, (V) $t = 1$ m, (VI) $t = 5$ m, (VII) 30 m, (VIII) $t = 2$ hr.

3.4. Energy absorption

Lattice structures are valued because of their combination of low weight and high energy absorption capabilities. One way to quantify this characteristic is through determining the specific energy absorption (SEA) parameter which expresses the energy-absorbing capability per unit mass of a structure. The SEA was determined for the double-T3 unit used herein from the quasi-static and dynamic tests using equation 1 shown below[62]:

$$SEA = \frac{EA}{m} = \frac{\int_0^{\delta} F d\delta}{m} \quad (1)$$

In this equation, F represents the compressive force, m signifies the mass, and EA corresponds to the total energy absorbed by the cellular structure, calculated as the area beneath the load-displacement curve up to the point of densification.

During compression, there are three main stages of deformation at which energy is absorbed: elastic, plastic, and densification[63]. The plastic stage has the highest energy absorption rate. Fortunately, ten-segurity structure, such as the T3, are known for having elongated plastic

regions, seen in Figs 6, 9, and 10 in which structures undergo significant deformation before failure or reaching densification. Fig. 13 shows the quasi-static SEA values from this study as blue squares and red circles for the double-T3 unit or lattices, respectively. Table 5 compares the SEA value for the single unit and lattice structure designs at quasi-static and dynamic strain rates with their associated average, maximum/minimum values. Interestingly, double-T3 unit has a lower SEA in the dynamic regime compared to the quasi-static regime while the lattice structure has a higher SEA in the dynamic regime compared to the quasi-static regime.

The observed ~20% reduction in the SEA of the Double-T3 unit under dynamic strain rates can be attributed to the increased stiffness and mass of the thicker bars and cables within the structure. Under dynamic loading, the higher mass of the thicker bars and cables within the unit can lead to increased inertial forces. This can cause stress concentrations that can result in higher peak forces, but do not necessarily lead to greater energy absorption. Instead, they can limit the structure's ability to deform plastically in a uniform manner, leading to earlier failure and less effective distribution of deformation energy. So, while higher inertial forces are present, the structure's ability to absorb

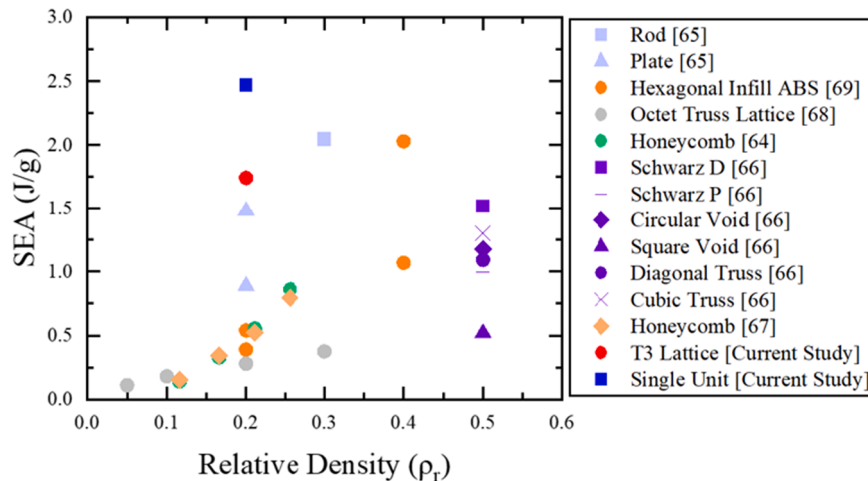


Fig. 13. SEA of similar lattice structures [64-69] compared with the structures used in the current study shown as red circles and blue squares.

energy efficiently is reduced due to early buckling and localized failure. This results in a lower overall SEA despite the presence of higher peak forces.

Additionally, the increased stiffness of the thicker elements reduces the structure's ability to redistribute stresses evenly, resulting in a more brittle response and limiting the plastic deformation that typically contributes to energy absorption. As a result, the SEA decreases, as the structure's ability to deform plastically is restricted under dynamic conditions. This restricted plastic deformation under dynamic loading leads to a less efficient energy absorption process compared to quasi-static conditions.

4. Conclusion

Herein, tensegrity-like structures composed of triangular tensegrity prisms were designed and fabricated using a vat photopolymerization additive manufacturing method. Both unit cell and lattice structures were manufactured. The structures were tested under quasi-static and dynamic loads and exhibited nonlinear responses that varied based on their geometries. Buckling of the members was the main source of failure of the structures under compressive loads. Microscopic analysis of the elements of the structure revealed fractures mostly occurred near nodal intersection. It is less likely that the surface imperfections were affecting the fracture initiation during testing. No significant strain rate dependency was observed in their mechanical responses. Under dynamic loads, the structures showed higher localized deformation compared to the quasi-static tests. Also, more significant instability was observed in the load-displacement graphs. This has been attributed to the stress wave traveling through the structure. The specific energy absorption was calculated for the structures in this study and compared with previously reported values. One of the main findings of this research was the variation of the specific energy absorption between the quasi-static and dynamic regimes. Interestingly, the double-T3 unit has a lower SEA in the dynamic regime compared to the quasi-static regime while the lattice structure has a higher SEA in the dynamic regime compared to the quasi-static regime. This indicates the crucial role that the geometry of the structure plays under different loading types in the energy absorption efficiency of tensegrity structures. Also, it was observed that these tensegrity-like structures were able to recover their shape after a significant deformation that adds a valuable dimension to their potential applications. The shape recovery weakened after subsequent compression, which is attributed to the failure of the members during each loading cycle. These structures offer opportunities for integrated and space structures due to their lightweight and compatibility. The results of this study contribute to the optimization of tensegrity structure design and geometries, providing valuable insights into the high strain rate behavior of these structures, which have not been previously studied in depth. Additional studies on the optimization of the structural design in order to obtain higher SEA values, additional shape recovery capability, and more stable behavior under high strain rate loading need to be conducted.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to rephrase some of the sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Keivan Davami: Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Russell Rowe:** Writing – original draft, Formal analysis, Data curation. **Ben Gullede:** Data curation. **Jesse Park:** Data curation. **Ali Beheshti:** Writing – original

draft. **Anthony Palazotto:** Investigation. **Fariborz Tavangarian:** Investigation. **Sadie Beck:** Writing – original draft, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

K.D. acknowledges the support from The Central Analytical Facility (CAF) at the University of Alabama and Alabama Water Institute (AWI).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijimpeng.2024.105208](https://doi.org/10.1016/j.ijimpeng.2024.105208).

Data availability

Data will be made available on request.

References

- [1] Schaedler TA, et al. Ultralight metallic microlattices. *Science* (1979) 2011;334(6058):962–5. <https://doi.org/10.1126/science.1211649>.
- [2] Meza LR, Das S, Greer JR. Strong, lightweight, and recoverable three-dimensional ceramic nanolattices. *Science* (1979) 2014;345(6202):1322–6. <https://doi.org/10.1126/science.1255908>.
- [3] Zheng X, et al. Ultralight, ultrastiff mechanical metamaterials. *Science* (1979) 2014;344(6190):1373–7. <https://doi.org/10.1126/science.1252291>.
- [4] Han SC, Lee JW, Kang K. A new type of low density material: shellular. *Adv Mater* 2015;27(37):5506–11. <https://doi.org/10.1002/adma.201501546>.
- [5] Mohsenizadeh M, Gasbarri F, Munther M, Beheshti A, Davami K. Additively-manufactured lightweight Metamaterials for energy absorption. *Mater Des* 2018; 139. <https://doi.org/10.1016/j.matdes.2017.11.037>.
- [6] Davami K, Mohsenizadeh M, Munther M, Palma T, Beheshti A, Momeni K. Dynamic energy absorption characteristics of additivelymanufactured shape-recovering lattice structures. *Mater Res Express* 2019;6(4). <https://doi.org/10.1088/2053-1591/aaf78c>.
- [7] Maldovan M. Sound and heat revolutions in phononics. *Nature* 2013;503(7475): 209–17. <https://doi.org/10.1038/nature12608>.
- [8] Suhartati T. *Soft Acoustic Metamaterials*. 342:106;2013. آب و خاک. May.
- [9] Cai C, Zhou J, Wu L, Wang K, Xu D, Ouyang H. Design and numerical validation of quasi-zero-stiffness metamaterials for very low-frequency band gaps. *Compos Struct* 2019;236:111862. <https://doi.org/10.1016/j.compstruct.2020.111862>.
- [10] Adhikari S, Mukhopadhyay T, Shaw A, Lavery NP. Apparent negative values of Young's moduli of lattice materials under dynamic conditions. *Int J Eng Sci* 2020; 150:103231. <https://doi.org/10.1016/j.ijengsci.2020.103231>.
- [11] Liu XN, Hu GK, Huang GL, Sun CT. An elastic metamaterial with simultaneously negative mass density and bulk modulus. *Appl Phys Lett* 2011;98(25):2009–12. <https://doi.org/10.1063/1.3597651>.
- [12] Fang N, et al. Ultrasonic metamaterials with negative modulus. *Nat Mater* 2006;5(6):452–6. <https://doi.org/10.1038/nmat1644>.
- [13] Hewage TAM, Alderson KL, Alderson A, Scarpa F. Double-Negative mechanical metamaterials displaying simultaneous negative stiffness and negative poisson's ratio properties. *Adv Mater* 2016;28(46):10323–32. <https://doi.org/10.1002/adma.201603959>.
- [14] Wang K, Chang YH, Chen YW, Zhang C, Wang B. Designable dual-material auxetic metamaterials using three-dimensional printing. *Mater Des* 2015;67:159–64. <https://doi.org/10.1016/j.matdes.2014.11.033>.
- [15] Zhao L, et al. Ultrasound beam steering with flattened acoustic metamaterial Luneburg lens. *Appl Phys Lett* 2020;116(7). <https://doi.org/10.1063/1.5140467>.
- [16] Ma G, Sheng P. Acoustic metamaterials: from local resonances to broad horizons. *Sci Adv* 2016;2(2). <https://doi.org/10.1126/sciadv.1501595>.
- [17] Yang P, Wu J, Zhao R, Han J. Research on local sound field control technology based on acoustic metamaterial triode structure. *Crystals* (Basel) 2020;10(3). <https://doi.org/10.3390/cryst10030204>.
- [18] Wang YF, Wang TT, Liu JP, Wang YS, Laude V. Guiding and splitting Lamb waves in coupled-resonator elastic waveguides. *Compos Struct* 2018;206:588–93. <https://doi.org/10.1016/j.compstruct.2018.08.088>.
- [19] Davami K, et al. Ultralight shape-recovering plate mechanical metamaterials. *Nat Commun* 2015;6:1–7. <https://doi.org/10.1038/ncomms10019>.
- [20] Intrigila C, Micheletti A, Nodargi NA, Artioli E, Bisegna P. Fabrication and experimental characterisation of a bistable tensegrity-like unit for lattice

- metamaterials. *Addit Manuf* 2022;57(June):102946. <https://doi.org/10.1016/j.addma.2022.102946>.
- [21] Bel Hadj Ali N, Smith IFC. Dynamic behavior and vibration control of a tensegrity structure. *Int J Solids Struct* 2010;47(9):1285–96. <https://doi.org/10.1016/j.ijsolstr.2010.01.012>.
- [22] Sultan C, Skelton R. Deployment of tensegrity structures. *Int J Solids Struct* 2003;40(18):4637–57. [https://doi.org/10.1016/S0020-7683\(03\)00267-1](https://doi.org/10.1016/S0020-7683(03)00267-1).
- [23] Chen M, Fraddosio A, Micheletti A, Pavone G, Piccioni MD, Skelton RE. Energy-efficient cable-actuation strategies of the V-Expander tensegrity structure subjected to five shape changes. *Mech Res Commun* 2022;127:104026. <https://doi.org/10.1016/j.mechrescom.2022.104026>.
- [24] R. Motro, *Tensegrity: structural systems for the future*. 2003.
- [25] J.Y. Zhang and M. Ohsaki, *Tensegrity Structures Form, Stability, and Symmetry*. 2015. doi: 10.1007/978-4-431-54813-3_1.
- [26] Y. Hurmuzlu and O. Nwokah, *The Mechanical Systems Design Handbook*. 2002. [Online]. Available: <https://doi.org/10.1201/9781420036749>.
- [27] The Georgia Dome and beyond: achieving lightweight-longspan structures. In: *Proceedings of the IASS-ASCE International Symposium*; 1994. p. 560–2.
- [28] "Analysis of the Georgia Dome cable roof," *Computing in Civil Engineering and Geographic Information Systems Symposium*, pp. 566–573.
- [29] Fraternali F, Santos F. Mechanical modeling of superelastic tensegrity braces for earthquake-proof structures. *Extreme Mech Lett* 2019;33:100578. <https://doi.org/10.1016/j.eml.2019.100578>.
- [30] Hirai S, Koizumi Y, Shibata M, Wang M, Bin L. Active shaping of a tensegrity robot via pre-pressure. In: 2013 IEEE/ASME International Conference on Advanced Intelligent Mechanisms: Mechanisms for Human Wellbeing, AIM 2013; 2013. p. 19–25. <https://doi.org/10.1109/AIM.2013.6584062>.
- [31] Shea K, Fest E, Smith IFC. Developing intelligent tensegrity structures with stochastic search. *Adv Eng Inf* 2002;16(1):21–40. [https://doi.org/10.1016/S1474-0346\(02\)00003-4](https://doi.org/10.1016/S1474-0346(02)00003-4).
- [32] Liu Y, Bi Q, Yue X, Wu J, Yang B, Li Y. A review on tensegrity structures-based robots. *Mech Mach Theory* 2022;168(October 2021):104571. <https://doi.org/10.1016/j.mechmachtheory.2021.104571>.
- [33] Zappetti D, Arandes R, Ajanic E, Floreano D. Variable-stiffness tensegrity spine. *Smart Mater Struct* 2020;29(7). <https://doi.org/10.1088/1361-665X/ab87e0>.
- [34] Tibert AG, Pellegrino S. Review of form-finding methods for tensegrity structures. *Int J Space Struct* 2011;26(3):241–55. <https://doi.org/10.1260/0266-3511.26.3.241>.
- [35] Connelly R, Terrel M. Globally rigid symmetric tensegrities. *Struct Topol* 1995;21(February):59–77.
- [36] S. Pellegrino, "Mechanics of kinematically indeterminate structures Dissertation submitted to the University of Cambridge for the Degree of Doctor of Philosophy by Sergio Pellegrino Corpus Christi College," no. 1986, 2016.
- [37] Kim K, Agogino AK, Toghiani A, Moon D, Taneja L, Agogino AM. Robust learning of tensegrity robot control for locomotion through form-finding. *IEEE Int Conf Intell Rob Syst* 2015;2015-Decem:5824–31. <https://doi.org/10.1109/IROS.2015.7354204>.
- [38] Kim K, Moon D, Bin JY, Agogino AM. Design of a spherical tensegrity robot for dynamic locomotion. *IEEE Int Conf Intell Rob Syst* 2017;2017-Sept:450–5. <https://doi.org/10.1109/IROS.2017.8202192>.
- [39] Pagitz M, Mirats Tur JM. Finite element based form-finding algorithm for tensegrity structures. *Int J Solids Struct* 2009;46(17):3235–40. <https://doi.org/10.1016/j.ijsolstr.2009.04.018>.
- [40] Luo A, Skelton RE, Liu H, Liu R, Guo H, Wang L. Structure of the ball tensegrity robot. 2014 IEEE Int Conf Rob Biomimetics, IEEE ROBIO 2014 2014;12(2):1781–6. <https://doi.org/10.1109/ROBIO.2014.7090593>.
- [41] Xue Y, Luo Y, Xu X. Form-finding of cable-strut structures with given cable forces and strut lengths. *Mech Res Commun* 2020;106:103530. <https://doi.org/10.1016/j.mechrescom.2020.103530>.
- [42] Moored KW, Bart-Smith H. The analysis of tensegrity structures for the design of a morphing wing. *J Appl Mech, Trans ASME* 2007;74(4):668–76. <https://doi.org/10.1115/1.2424718>.
- [43] Vangelatos Z, Micheletti A, Grigoropoulos CP, Fraternali F. Design and testing of bistable lattices with tensegrity architecture and nanoscale features fabricated by multiphoton lithography. *Nanomaterials* 2020;10(4). <https://doi.org/10.3390/nano10040652>.
- [44] Pajunen K, Johans P, Pal RK, Rimoli JJ, Daraio C. Design and impact response of 3D-printable tensegrity-inspired structures. *Mater Des* 2019;182:107966. <https://doi.org/10.1016/j.matdes.2019.107966>.
- [45] Pajunen K, Celli P, Daraio C. Prestrain-induced bandgap tuning in 3D-printed tensegrity-inspired lattice structures. *Extreme Mech Lett* 2021;44:101236. <https://doi.org/10.1016/j.eml.2021.101236>.
- [46] Bauer J, Kraus JA, Crook C, Rimoli JJ, Valdevit L. Tensegrity metamaterials: toward failure-resistant engineering systems through delocalized deformation. *Adv Mater* 2021;33(10):1–9. <https://doi.org/10.1002/adma.202005647>.
- [47] Furqan M, Naushad Alam M. Finite element analysis of a Stewart Platform using flexible joints. 1st Int 16th Natl Conf Mach Mech, iNaCoMM 2013 2013:1044–9.
- [48] Lobontiu N, Paine JSN, Garcia E, Goldfarb M. Corner-ribbed flexure hinges. *J Mech Des, Trans ASME* 2001;123(3):346–52. <https://doi.org/10.1115/1.1372190>.
- [49] Amendola A, Favata A, Micheletti A. On the mechanical modeling of tensegrity columns subject to impact loading. *Front Mater* 2018;5(April):1–8. <https://doi.org/10.3389/fmats.2018.00022>.
- [50] Amendola A, Hernández-Nava E, Goodall R, Todd I, Skelton RE, Fraternali F. On the additive manufacturing, post-tensioning and testing of bi-material tensegrity structures. *Compos Struct* 2015;131:66–71. <https://doi.org/10.1016/j.compstruct.2015.04.038>.
- [51] Fraternali F, Carpentieri G, Amendola A. On the mechanical modeling of the extreme softening/stiffening response of axially loaded tensegrity prisms. *J Mech Phys Solids* 2015;74:136–57. <https://doi.org/10.1016/j.jmps.2014.10.010>.
- [52] Davini C, Micheletti A, Podio-Guidugli P. On the impulsive dynamics of T3 tensegrity chains. *Meccanica* 2016;51(11):2763–76. <https://doi.org/10.1007/s11012-016-0495-y>.
- [53] Oppenheim IJ, Williams WO. Mechanics of Tensegrity Prisms. *ISARC Proc 15th Int Symp Autom Rob Construct* 2017:473–9. <https://doi.org/10.22260/isarc1998/0034>.
- [54] Cai J, Wang X, Yang R, Feng J. Mechanical behavior of tensegrity structures with High-mode imperfections. *Mech Res Commun* 2018;94:58–63. <https://doi.org/10.1016/j.mechrescom.2018.09.006>.
- [55] Rimoli JJ, Pal RK. Mechanical response of 3-dimensional tensegrity lattices. *Compos B Eng* 2017;115:30–42. <https://doi.org/10.1016/j.compositesb.2016.10.046>.
- [56] He SS, Yin X, Zhang LY, Gao ZY, Xu GK. Directional snapping instability in a bistable tensegrity under uniaxial loads. *Compos Struct* 2022;283(December 2021):115153. <https://doi.org/10.1016/j.compstruct.2021.115153>.
- [57] Zhang J, Ohsaki M, Rimoli JJ, Kogiso K. Optimization for energy absorption of 3-dimensional tensegrity lattice with truncated octahedral units. *Compos Struct* 2021;267(February):113903. <https://doi.org/10.1016/j.compstruct.2021.113903>.
- [58] Micheletti A, Ruscica G, Fraternali F. On the compact wave dynamics of tensegrity beams in multiple dimensions. *Nonlinear Dyn* 2019;98(4):2737–53. <https://doi.org/10.1007/s11071-019-04986-8>.
- [59] Fraternali F, Carpentieri G, Amendola A, Skelton RE, Nesterenko VF. Multiscale tunability of solitary wave dynamics in tensegrity metamaterials. *Appl Phys Lett* 2014;105(20). <https://doi.org/10.1063/1.4902071>.
- [60] R.E. Skelton and M.C. De Oliveira, *Skelton-DeOliveira2009 Book TensegritySystems*.
- [61] Rimoli JJ. On the impact tolerance of tensegrity-based planetary landers. 57th AIAA/ASCE/AHS/ASC Struct, Struct Dyn, Mater Conf 2016;(January). <https://doi.org/10.2514/6.2016-1511>.
- [62] Cao X, Duan S, Liang J, Wen W, Fang D. Mechanical properties of an improved 3D-printed rhombic dodecahedron stainless steel lattice structure of variable cross section. *Int J Mech Sci* 2018;145. <https://doi.org/10.1016/j.ijmecsci.2018.07.006>.
- [63] Gibson LJ, Editor G. *Cellular Solids*. *Mater Res Soc* 2003;2021(APRIL):270–4.
- [64] Bates SRG, Farrow IR, Trask RS. 3D printed polyurethane honeycombs for repeated tailored energy absorption. *Mater Des* 2016;112:172–83. <https://doi.org/10.1016/j.matdes.2016.08.062>.
- [65] J.S. Weeks, "Mechanical Response of Lattice Structures under High Strain-Rate and Shock Loading," p. 128, 2023.
- [66] Niknam H, Akbarzadeh AH. Graded lattice structures: simultaneous enhancement in stiffness and energy absorption. *Mater Des* 2020;196:109129. <https://doi.org/10.1016/j.matdes.2020.109129>.
- [67] Habib FN, Iovenitti P, Masood SH, Nikzad M. In-plane energy absorption evaluation of 3D printed polymeric honeycombs. *Virtual Phys Prototyp* 2017;12(2):117–31. <https://doi.org/10.1080/17452759.2017.1291354>.
- [68] Tancogne-Dejean T, Spierings AB, Mohr D. Additively-manufactured metallic micro-lattice materials for high specific energy absorption under static and dynamic loading. *Acta Mater* 2016;116:14–28. <https://doi.org/10.1016/j.actamat.2016.05.054>.
- [69] Dave HK, Karumuri RT, Prajapati AR, Rajpurohit SR. Specific energy absorption during compression testing of ABS and FPU parts fabricated using LCD-SLA based 3D printer. *Rapid Prototyp J* 2022;28(8):1530–40. <https://doi.org/10.1108/RPJ-04-2021-0075>.